SSC MTS 2024 question paper with answers

30 October 2024 · Shift 1

79 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 11 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 16 questions

Q.1

S gets 78, 75, 68, 64 and 58 marks in Hindi, English, Chemistry, Physics and Mathematics, respectively. What are his average marks per subject?

(A) 66.6
(B) 68.6
(C) 68.8
(D) 66.8

Q.2

In a group of 200 members, 45% are teenagers and the remaining are adults. The average weight of the whole group is 62 kg. If the average weight of teenagers is 56 kg, then the nearest integral value of the average weight of adults is _____.

(A) 79 kg
(B) 67 kg
(C) 84 kg
(D) 65 kg

Q.3

The average of the first twelve multiples of 11 is:

(A) 68.5
(B) 70.5
(C) 71.5
(D) 69.5

Q.5

The diameter of a hollow hemisphere is 40 cm. Water is filled in it up to its brim. When spherical lead shots each of diameter 0.8 cm are dropped into the hemisphere, three-fifths of the water flows out. How many lead shots were dropped into the hemisphere?

(A) 37500
(B) 35500
(C) 30500
(D) 35700

Q.7

The HCF of 21 and 49 is:

(A) 7
(B) 3
(C) 2
(D) 1

Q.8

What will be the average of all the odd numbers from 11 to 71 (both inclusive)?

(A) 39
(B) 41
(C) 47
(D) 31

Q.10

The difference between the selling prices with 19% profit and 11% loss is Rs.454.50. What is the cost price of the item?

(A) Rs.1,525
(B) Rs.1,515
(C) Rs.1,520
(D) Rs.1,505

Q.11

If a man takes 36 minutes to cover a certain distance at a speed of 6 km/h, then the distance covered by him is:

(A) 2.5 km
(B) 2 km
(C) 3.6 km
(D) 3 km

Q.13

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 5% per year rate of interest for 4 years.

(A) 460
(B) 440
(C) 540
(D) 500

Q.14

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 5% per year rate of interest for 4 years.

(A) 600
(B) 540
(C) 640
(D) 560

Q.15

A person spends 75% of his income. If his income increases by 15% and expenditure decreases by 10%, then what percentage increase will occur in his savings?

(A) 90%
(B) 67.5%
(C) 22.5%
(D) 47.5%

Q.16

Gagandeep marks his goods 50% above the cost price and offers a discount of 25% on the marked price to his customers. What is the percentage gain of Gagandeep?

(A) 13.5%
(B) 12.5%
(C) 13.0%
(D) 12.0%

Q.17

A cooperative society increases the annual rate of simple interest from 10% to 16½ %, then the yearly income of a person increases by ₹1,077.70. The simple interest (in ₹) on the sum at 10% for four years will be:

(A) 9,120.3
(B) 6,632
(C) 16,580
(D) 8,240

Q.18

A speed of 16 km/h is the same as:(Up to 1 place of decimal)

(A) 4.4 m/sec
(B) 8 m/sec
(C) 2 m/sec
(D) 8.4 m/sec

Q.19

By selling an article for ₹19.5, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 30%?

(A) 6.55
(B) 3.55
(C) 4.55
(D) 5.55

Q.20

The weight of a sphere is jointly proportional to the cube of its radius and the density of the material of which it is made. The radii of two spheres are 17 : 8 and the densities of materials 3 : 4. If the weight of the second sphere be 40 kg, then the approximate weight of the first sphere is:

(A) 287.87 kg
(B) 96 kg
(C) 230 kg
(D) 123.26 kg

Reasoning Ability and Problem Solving 13 questions

Q.2

NUQM is related to PWSO in a certain way based on the English alphabetical order. In the same way, JQMI is related to LSOK. To which of the following is FMIE related, following the same logic?

(A) HOKG
(B) OHKG
(C) HOGK
(D) OHGK

Q.3

What should come in place of the question mark (?) in the given series based on the English alphabetical order? NRT, JNP, FJL, BFH, ?

(A) XZS
(B) XDF
(C) XAD
(D) XBD

Q.6

S, T, U, V, W, X and Y are sitting around a circular table facing the centre (but not necessarily in the same order). Only four people are sitting between S and W when counted from the right of S. Only one person is sitting between W and Y when counted from the left of W. Only two people are sitting between S and X when counted from the left of S. Only two people are sitting between U and Y when counted from the left of U . Only one person is sitting between U and V when counted from the left of V. Who is sitting to the immediate right of X ?

(A) W
(B) S
(C) V
(D) Y

Q.7

What will come in the place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 12 ÷ 13 + 128 – 343 × 49 = ?

(A) 45
(B) 53
(C) 54
(D) 35

Q.8

What should come in place of the question mark (?) in the following series? -553, -532, -511, -490, -469, ?

(A) 458
(B) -448
(C) 448
(D) -458

Q.12

28 is related to 168 following a certain logic. Following the same logic, 39 is related to 234. To which of the following is 43 related following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 215
(B) 258
(C) 222
(D) 217

Q.13

What will come in the place of '?' in the following equation, if '+' and '÷' are interchanged and '×' and '–' are interchanged? 24 + 3 ÷ 8 × 2 – 6 =?

(A) 6
(B) – 6
(C) 4
(D) 18.75

Q.14

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some rhinos are birds. All penguins are birds. Conclusions: (I): Some rhinos are penguins. (II): All birds are penguins.

(A) Both conclusions (I) and (II) follow.
(B) Only conclusion (I) follows.
(C) Only conclusion (II) follows.
(D) Neither conclusion (I) nor (II) follows.

Q.15

U, V, W, X, Y, Z and A are sitting around a circular table facing the centre (but not necessarily in the same order). X sits third to the left of Y. W sits second to the left of V. Only Y sits between A and W. U is not an immediate neighbour of X. Who sits fourth to the right of Z?

(A) W
(B) A
(C) Y
(D) U

Q.16

In a certain code language, 'RESHMA' is coded as 'XJEPBO' and 'PRITHAM' is coded as 'JXEQFOM'. How will 'GOURI' be coded in that language?

(A) FORLC
(B) FORLD
(C) FDRLD
(D) EORLD

Q.17

FLLC is related to HNNE in a certain way based on the English alphabetical order. In the same way, AFFW is related to CHHY. To which of the following is PVVM related, following the same logic?

(A) ROXX
(B) XXRO
(C) RXXO
(D) XXOR

Q.18

In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 7, 27; 11, 43

(A) 15, 59
(B) 14, 58
(C) 14, 59
(D) 15, 58

Q.19

In a certain code language, 'LIME' is coded as '4826' and 'FOLD' is coded as '9378'. What is the code for 'L' in that language?

(A) 2
(B) 7
(C) 4
(D) 8

General Awareness 25 questions

Q.1
As per the 2011 census, which of the following union territories had the lowest population?

(A) Andaman and Nicobar Islands
(B) Puducherry
(C) Lakshadweep
(D) Chandigarh

Q.2
What is the study of the interactions between living organisms and their environment called?

(A) Chemistry
(B) Ecology
(C) Geology
(D) Biology

Q.3
In which of the following states is the Mamallapuram Utsav celebrated?

(A) Kerala
(B) Karnataka
(C) Tamil Nadu
(D) Uttarakhand

Q.4
Which of the following is NOT considered as a Fundamental Duty?

(A) To develop the scientific temper, humanism and the spirit of inquiry and reform
(B) To value and preserve the rich heritage of our composite culture
(C) To uphold and protect the sovereignty, unity and integrity of India
(D) To uphold and protect secularism

Q.5
Which Kuchipudi dancer received National Nritya Shiromani Award 2022?

(A) Anindita Neogy Anaam
(B) Jatin Goswami
(C) Aparna Satheesan
(D) Anita Sharma

Q.6
Which of the following rivers flows north of the Tropic of Cancer in India?

(A) Tapi
(B) Mahanadi
(C) Penganga
(D) Yamuna

Q.7
In May 2023, who was sworn in as the new Central Vigilance Commissioner of India?

(A) GC Murmu
(B) PK Srivastava
(C) Amit Khare
(D) Suresh N Patel

Q.8
The Huttari dance, observed as an annual tradition, is a leisurely performance that takes place during the harvesting season in _____.

(A) Karnataka
(B) Odisha
(C) Telangana
(D) Andhra Pradesh

Q.9
Scissor style and Western Roll are the techniques used in:

(A) high jump
(B) long jump
(C) pole vault
(D) triple jump

Q.10
Match the following parties with their reserved symbols and choose the INCORRECT one among the following:

(A) Bahujan Samaj Party – Elephant
(B) All India Trinamool Congress – Sickle
(C) National People's Party – Book
(D) Nationalist Congress party – Clock

Q.11
Large tidal waves caused by an underwater earthquake or volcanic eruption are known as a _____.

(A) tsunami
(B) tornado
(C) hurricane
(D) typhoon

Q.12
Which of the following National Waterways (NW) is associated with the Udyogamandal and the Champakkara canals?

(A) NW-3
(B) NW-5
(C) NW-25
(D) NW-10

Q.13
Most of the Harappan sites are primarily divided into _____ parts.

(A) eight
(B) two
(C) six
(D) four

Q.14
On 19 July 2023, who was appointed as the 25th Director General of the Indian Coast Guard?

(A) BS Raju
(B) VK Singh
(C) Rakesh Pal
(D) K Natarajan

Q.15
With which of the following organisations was Henry Louis Vivian Derozio, a teacher of Hindu College of Calcutta, associated?

(A) Paramahansa Mandali
(B) Young Bengal Movement
(C) Sree Narayana Dharma Paripalana Yogam
(D) Rahnumai Mazdayasnan Sabha

Q.16
Meher had two large chunks of coal. She set fire to the first chunk and broke the second one into pieces. Help Meher understand what kind of changes happened to the two chunks of coal at the end of the activity by choosing all the applicable statements. A. Both chunks of coal underwent physical changes only. B. The changes were reversible for the first chunk of coal only. C. Only physical changes took place for the second chunk of coal. D. Both chunks of coal underwent chemical changes only. E. The first case is an example of chemical changes.

(A) Only D
(B) Only A and B
(C) Only C and E
(D) Only B and D

Q.17
With reference to provinces under Mauryan dynasty, which of the following pairs is correct?

(A) Prachyapatha (East) - Toshali
(B) Dakshinapatha (South) - Ujjain
(C) Uttarapatha (North) - Suvarnagiri
(D) Avantiratha (West) - Taxila

Q.18
As a part of the Non-Cooperation Movement in 1922, the angry peasants had set ablaze the police station of which place?

(A) Dandi
(B) Chauri Chaura
(C) Kakori
(D) Dhaka

Q.19
Gangubai Hangal, a renowned Indian music personality, is recognised for her contributions in which of the following fields?

(A) Ghazal Singing
(B) Rabindra Sangeet
(C) Sufi Music
(D) Hindustani Classical Music

Q.20
What do you call the body cavity that is lined by mesoderm?

(A) Coelom
(B) Diploblasts
(C) Radial symmetry
(D) Triploblasts

Q.21
The God of Small Things is a novel authored by whom among the following authors?

(A) Shashi Tharoor
(B) Fakira
(C) Arundhati Roy
(D) Rohinton Mistry

Q.22
In which of the following empires of South India was the Kamalapuram tank built in the 15th century?

(A) Bidar
(B) Chola
(C) Vijayanagara
(D) Bijapur

Q.23
If labour is employed in a job but is not actually utilised for the production of goods and services, it is known as:

(A) disguised unemployment
(B) educated unemployment
(C) urban unemployment
(D) seasonal unemployment

Q.24
In basketball, how many points are awarded when a basket is scored from outside the 6.25-metre line?

(A) Six points
(B) Five points
(C) Two points
(D) Three points

Q.25
What is the role of a commercial bank in the economy?

(A) To provide financial services, such as accepting deposits, lending money and facilitating payments
(B) To print and issue currency
(C) To set interest rates for the entire banking system
(D) To regulate the money supply

English Language and Comprehension 25 questions

Q.1
Select the option with the correctly spelt word to fill in the blank. The delta of river _____ looks like a bird's foot because a lot of sediment is carried into the calm sea water.

(A) Mississippi
(B) Mississipi
(C) Mississipppi
(D) Missisippi

Q.2
Select the option that can be used as a one-word substitute for the given group of words. The irrational fear of snakes

(A) Xenophobia
(B) Ophidiophobia
(C) Necrophobia
(D) Nomophobia

Q.3

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in the blanks. Cuba is known as the sugar bowl of the world. It is part of the Caribbean _____. This country _____ a lot of sugar in the decade of 1960s.

(A) sector; created
(B) area; cropped
(C) region; produced
(D) stretch; processed

Q.4

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Despite the challenges, / the team is determined to / completed / the project on time and within budget.

(A) Despite the challenges
(B) the team is determined to
(C) completed
(D) the project on time and within budget

Q.5

Select the option that correctly rectifies the underlined spelling error in the given sentence. In the first half century after U.S. independence, a major “preportion” of the nation's labour force shifted from the agricultural to the manufacturing sector.

(A) proportion
(B) priposon
(C) prepotion
(D) preposion

Q.6

Select the grammatically correct sentence from the given options.

(A) The responsibility was given to each student to keep quiet during the seminar.
(B) The responsibility was given to each student to keep quieter during the seminar.
(C) The responsibility is given to each student to keep quieter during the seminar.
(D) The responsibility was give to each student to keep quieter during the seminar.

Q.7

Select the most appropriate meaning of the underlined word in the context of the following sentence. We should go to the Himalayas this “spring”.

(A) A season
(B) Coiled metal
(C) A water body
(D) To jump

Q.8

Select the INCORRECTLY spelt word.

(A) Assesment
(B) Exaggeration
(C) Committee
(D) Sacrilege

Q.9

The given sentence is divided into four segments. Select the segment with the grammatical error. I was surprised / to see my / daughter, Manya, riding her / new bicycle confident.

(A) I was surprised
(B) daughter, Manya, riding her
(C) to see my
(D) new bicycle confident

Q.10

Select the sentence that has a grammatical error.

(A) We came in the meeting room quietly.
(B) He came across an interesting article while researching.
(C) The company came up with a brilliant marketing strategy.
(D) She always comes down with a cold in winter.

Q.11

Select the most appropriate option to fill in the blank. How long have you been _____ these contact lenses?

(A) swimming
(B) prevailing
(C) wearing
(D) seeing

Q.12

Select the most appropriate option to fill in the blank. The _____ model who can be selected for this advertisement is Meera Narayan.

(A) most best
(B) best
(C) more better
(D) most good

Q.13

Identify the preposition that needs to be omitted to correct the sentence. They live near to the park.

(A) live
(B) to
(C) near
(D) They

Q.14

select the most appropriate idiom/phrase to fill in the blank and complete the sentence. It was all _____. He did not really mean what he said.

(A) blue in the face
(B) tuck in
(C) tongue in cheek
(D) a storm in a tea cup

Q.15

Select the most appropriate ANTONYM of the given word. Ally

(A) Enemy
(B) Infamy
(C) Dreamy
(D) Gloomy

Q.16

Select the most appropriate synonym of the underlined word. Ritu is a “diligent” girl who looks after her aged parents.

(A) innocent
(B) stupid
(C) hardworking
(D) lethargic

Q.17

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select ‘No substitution required’. Manisha “set apart” with her friends to explore the countryside.

(A) set aside
(B) set in
(C) set out
(D) No substitution required

Q.18

Select the most appropriate meaning of the underlined word in the given sentence. The military operation was part of a “conspiracy” against national security.

(A) willfulness
(B) Motivation
(C) Enthusiasm
(D) Plot

Q.19

Select the most appropriate meaning of the underlined word in the given sentence. I “agreed” with your assessment.

(A) Disagreed
(B) Rejected
(C) Doubted
(D) Supported

Q.20

Select the most appropriate idiom to replace the underlined segment in the given sentence. We were asked to vote for the candidates by “raising our hands to indicate voting for or against a proposition”.

(A) getting out of hand
(B) changing hands
(C) a show of hands
(D) tipping our hands

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A weed is an untamed plant that (1)_____ where it is not wanted, usually among gardens or crops. The majority of weeds are (2)_____. While some are toxic to horses and cattle, others harm mature crops by (3)_____them of water, sunlight, soil and minerals. There are various ways that weeds (4)_____. They are (5)_____ transported from one location to another as trash, dust or fodder.

Select the most appropriate option to fill in blank number 1.

(A) amends
(B) spreads
(C) perfects
(D) evolves

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A weed is an untamed plant that (1)_____ where it is not wanted, usually among gardens or crops. The majority of weeds are (2)_____. While some are toxic to horses and cattle, others harm mature crops by (3)_____them of water, sunlight, soil and minerals. There are various ways that weeds (4)_____. They are (5)_____ transported from one location to another as trash, dust or fodder.

Select the most appropriate option to fill in blank number 2.

(A) delicates
(B) grace
(C) dangerous
(D) vices

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A weed is an untamed plant that (1)_____ where it is not wanted, usually among gardens or crops. The majority of weeds are (2)_____. While some are toxic to horses and cattle, others harm mature crops by (3)_____them of water, sunlight, soil and minerals. There are various ways that weeds (4)_____. They are (5)_____ transported from one location to another as trash, dust or fodder.

Select the most appropriate option to fill in blank number 3.

(A) digesting
(B) relieving
(C) spoiling
(D) depriving

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A weed is an untamed plant that (1)_____ where it is not wanted, usually among gardens or crops. The majority of weeds are (2)_____. While some are toxic to horses and cattle, others harm mature crops by (3)_____them of water, sunlight, soil and minerals. There are various ways that weeds (4)_____. They are (5)_____ transported from one location to another as trash, dust or fodder.

Select the most appropriate option to fill in blank number 4.

(A) propagate
(B) extension
(C) entertain
(D) breach

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A weed is an untamed plant that (1)_____ where it is not wanted, usually among gardens or crops. The majority of weeds are (2)_____. While some are toxic to horses and cattle, others harm mature crops by (3)_____them of water, sunlight, soil and minerals. There are various ways that weeds (4)_____. They are (5)_____ transported from one location to another as trash, dust or fodder.

Select the most appropriate option to fill in blank number 5.

(A) rare

(B) consequence

(C) every

(D) frequently

Answer key

Numerical and Mathematical Ability

Q.1 **(B)**	Q.2 **(B)**	Q.3 **(C)**	Q.5 **(A)**	Q.7 **(A)**	Q.8 **(B)**
Q.10 **(B)**	Q.11 **(C)**	Q.13 **(D)**	Q.14 **(A)**	Q.15 **(A)**	Q.16 **(B)**
Q.17 **(B)**	Q.18 **(A)**	Q.19 **(B)**	Q.20 **(A)**		

Reasoning Ability and Problem Solving

Q.2 **(A)**	Q.3 **(D)**	Q.6 **(A)**	Q.7 **(D)**	Q.8 **(B)**	Q.12 **(B)**
Q.13 **(C)**	Q.14 **(D)**	Q.15 **(D)**	Q.16 **(B)**	Q.17 **(C)**	Q.18 **(A)**
Q.19 **(D)**					

General Awareness

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(C)**	Q.4 **(D)**	Q.5 **(C)**	Q.6 **(D)**
Q.7 **(B)**	Q.8 **(A)**	Q.9 **(A)**	Q.10 **(B)**	Q.11 **(A)**	Q.12 **(A)**
Q.13 **(B)**	Q.14 **(C)**	Q.15 **(B)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(B)**
Q.19 **(D)**	Q.20 **(A)**	Q.21 **(C)**	Q.22 **(C)**	Q.23 **(A)**	Q.24 **(D)**
Q.25 **(A)**					

English Language and Comprehension

Q.1 **(A)**	Q.2 **(B)**	Q.3 **(C)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(A)**
Q.7 **(A)**	Q.8 **(A)**	Q.9 **(D)**	Q.10 **(A)**	Q.11 **(C)**	Q.12 **(B)**
Q.13 **(B)**	Q.14 **(C)**	Q.15 **(A)**	Q.16 **(C)**	Q.17 **(C)**	Q.18 **(D)**
Q.19 **(D)**	Q.20 **(C)**	Q.21 **(B)**	Q.22 **(C)**	Q.23 **(D)**	Q.24 **(A)**
Q.25 **(D)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (B) 68.6

78 + 75 + 68 + 64 + 58 = 343, and 343 ÷ 5 = 68.6.

Q.2 · Answer (B) 67 kg

The whole group weighs 200 × 62 = 12,400 kg and the 90 teenagers 90 × 56 = 5,040 kg, so the 110 adults weigh 7,360 kg: 7,360 ÷ 110 = 66.9, or 67 kg.

Q.3 · Answer (C) 71.5

The multiples 11, 22, …, 132 are evenly spaced, so their average is (11 + 132)/2 = 71.5.

Q.5 · Answer (A) 37500

The hemisphere (radius 20 cm) holds (2/3)π × 20^3 = 16,000π/3 cm^3, so the shots displace three-fifths of it, 3,200π cm^3. Each shot (radius 0.4 cm) has volume (4/3)π × 0.4^3 = 0.256π/3 cm^3, and 3,200π ÷ (0.256π/3) = 37,500.

Q.7 · Answer (A) 7

21 = 3 × 7 and 49 = 7 × 7, so the HCF is 7.

Q.8 · Answer (B) 41

The odd numbers 11, 13, …, 71 are evenly spaced, so their average is (11 + 71)/2 = 41.

Q.10 · Answer (B) Rs.1,515

The two selling prices differ by 19% + 11% = 30% of the cost price, so the cost price is 454.50 ÷ 0.3 = ₹1,515.

Q.11 · Answer (C) 3.6 km

36 minutes is 0.6 hour, and 6 × 0.6 = 3.6 km.

Q.13 · Answer (D) 500

SI = 2,500 × 5 × 4/100 = ₹500.

Q.14 · Answer (A) 600

SI = 3,000 × 5 × 4/100 = ₹600.

Q.15 · Answer (A) 90%

On an income of 100 he spends 75 and saves 25. Afterwards he earns 115 and spends 67.5, saving 47.5: a rise of 22.5 on 25, or 90%.

Q.16 · Answer (B) 12.5%

Selling price = 1.5 × 0.75 = 1.125 times the cost price: a gain of 12.5%.

Q.17 · Answer (B) 6,632

The rise of 16.5% − 10% = 6.5% of the sum is ₹1,077.70, so the sum is ₹16,580. SI at 10% for four years = 16,580 × 10 × 4/100 = ₹6,632.

Q.18 · Answer (A) 4.4 m/sec

1 km/h = 5/18 m/s, so 16 km/h = 16 × 5/18 = 4.44 m/s, or 4.4 m/s.

Q.19 · Answer (B) 3.55

The selling price must rise from 1.1 to 1.3 times the cost price, an increase of 19.5 × 0.2/1.1 = 3.5454…, or ₹3.55 to two decimal places.

Q.20 · Answer (A) 287.87 kg

W_1/W_2 = $(17/8)^3$ × 3/4 = (4,913/512) × 3/4 = 14,739/2,048, so the first sphere weighs 40 × 14,739/2,048 ≈ 287.87 kg.

Reasoning Ability and Problem Solving

Q.2 · Answer (A) HOKG
Each letter moves 2 places forward: N → P, U → W, Q → S, M → O, and J → L, Q → S, M → O, I → K. So FMIE → HOKG.

Q.3 · Answer (D) XBD
Each letter moves 4 places back every step (B wraps round to X): N, J, F, B, X; R, N, J, F, B; T, P, L, H, D. The next cluster is XBD.

Q.6 · Answer (A) W
Facing the centre, a person's right is anticlockwise and left clockwise. Clockwise from S the seats are S, U, W, X, Y, T, V: four sit between S and W on S's right, one between W and Y on W's left, two between S and X on S's left, two between U and Y on U's left and one between V and U on V's left. X's immediate right is W.

Q.7 · Answer (D) 35
After interchanging the signs: 12 × 13 – 128 + 343 ÷ 49 = 156 – 128 + 7 = 35.

Q.8 · Answer (B) -448
Each term is 21 more than the one before: –469 + 21 = –448.

Q.12 · Answer (B) 258
The second number is six times the first: 28 × 6 = 168 and 39 × 6 = 234. Likewise 43 × 6 = 258.

Q.13 · Answer (C) 4
After interchanging the signs: 24 ÷ 3 + 8 – 2 × 6 = 8 + 8 – 12 = 4.

Q.14 · Answer (D) Neither conclusion (I) nor (II) follows.
The rhinos that are birds need not be penguins, so I does not follow; and all penguins being birds does not make all birds penguins, so II does not follow.

Q.15 · Answer (D) U
Facing the centre, a person's left is clockwise and right anticlockwise. A and W sit either side of Y; W cannot be on Y's left, since V would then need A's seat. So clockwise from Y the seats are Y, A, Z, X, V, U, W (U, not next to X, takes the seat between V and W). Four places to Z's right is U.

Q.16 · Answer (B) FORLD
Each letter moves 3 places back and the result is reversed: RESHMA → OBPEJX → XJEPBO and PRITHAM → MOFQEXJ → JXEQFOM. So GOURI → DLROF → FORLD.

Q.17 · Answer (C) RXXO
Each letter moves 2 places forward: F → H, L → N, C → E, and A → C, F → H, W → Y. So PVVM → RXXO.

Q.18 · Answer (A) 15, 59
The second number is four times the first minus 1: 7 × 4 – 1 = 27 and 11 × 4 – 1 = 43. Likewise 15 × 4 – 1 = 59.

Q.19 · Answer (D) 8
L is the only letter in both words and 8 the only digit in both codes, so L = 8.

General Awareness

Q.1 · Answer (C) Lakshadweep
Lakshadweep, with 64,473 people in 2011, was the least populous Union Territory; Andaman and Nicobar had about 3.8 lakh, and Chandigarh and Puducherry over 10 lakh each.

Q.2 · Answer (B) Ecology
Ecology is the study of how living organisms interact with one another and with their surroundings.

Q.3 · Answer (C) Tamil Nadu
The Mamallapuram Dance Festival is held at Mamallapuram (Mahabalipuram) in Tamil Nadu, against the backdrop of the Pallava rock monuments.

Q.4 · Answer (D) To uphold and protect secularism
Article 51A asks citizens to uphold the sovereignty, unity and integrity of India, to value and preserve the heritage of our composite culture and to develop the scientific temper; protecting secularism is not among the listed duties.

Q.5 · Answer (C) Aparna Satheesan
SSC's key gives Aparna Satheesan. Of the others, Jatin Goswami is a Sattriya dancer and Anindita Neogy Anaam a Kathak dancer.

Q.6 · Answer (D) Yamuna
The Yamuna, which rises in Uttarakhand and joins the Ganga at Prayagraj, flows wholly north of the Tropic of Cancer (23.5° N). The Tapi, Mahanadi and Penganga flow south of it.

Q.7 · Answer (B) PK Srivastava
Praveen Kumar Srivastava, acting Central Vigilance Commissioner since December 2022, was sworn in as CVC by the President on 29 May 2023. Suresh N Patel was his predecessor.

Q.8 · Answer (A) Karnataka
Huttari is the harvest festival of the Kodavas of Kodagu (Coorg) in Karnataka, celebrated with their traditional dances.

Q.9 · Answer (A) high jump
The scissors and the western roll are ways of clearing the bar in the high jump, as is the Fosbury flop used today.

Q.10 · Answer (B) All India Trinamool Congress – Sickle
The All India Trinamool Congress's symbol is 'flowers and grass', not a sickle. The BSP's elephant, the NPP's book and the NCP's clock are correct.

Q.11 · Answer (A) tsunami
A tsunami is a series of huge sea waves set off by an undersea earthquake, volcanic eruption or landslide.

Q.12 · Answer (A) NW-3
National Waterway 3, the West Coast Canal in Kerala (Kollam to Kottapuram), includes the Champakara and Udyogamandal canals.

Q.13 · Answer (B) two
Harappan settlements such as Mohenjodaro were divided into two parts: a smaller but higher citadel and a much larger but lower town.

Q.14 · Answer (C) Rakesh Pal
On 19 July 2023 the Ministry of Defence announced Rakesh Pal's appointment as the 25th Director General of the Indian Coast Guard.

Q.15 · Answer (B) Young Bengal Movement
Henry Louis Vivian Derozio, who taught at Hindu College, inspired the radical Young Bengal Movement among his students.

Q.16 · Answer (C) Only C and E
Burning turns coal into new substances such as ash and carbon dioxide, a chemical change that cannot be reversed (E); breaking the second chunk only changes its size, a physical change (C). A, B and D are false.

Q.17 · Answer (A) Prachyapatha (East) - Toshali
Tosali (Toshali) was the capital of the Mauryas' eastern province, Kalinga. Taxila was the northern capital, Ujjain the western (Avanti) and Suvarnagiri the southern.

Q.18 · Answer (B) Chauri Chaura
In February 1922 a crowd at Chauri Chaura, in Gorakhpur district, set the police station on fire; Gandhi then called off the Non-Cooperation Movement.

Q.19 · Answer (D) Hindustani Classical Music
Gangubai Hangal was a Hindustani classical vocalist of the Kirana gharana.

Q.20 · Answer (A) Coelom
The coelom is the body cavity lined by mesoderm; animals that have one are called coelomates.

Q.21 · Answer (C) Arundhati Roy
'The God of Small Things' (1997), Arundhati Roy's first novel, won the Booker Prize.

Q.22 · Answer (C) Vijayanagara
The Kamalapuram tank, built in the early fifteenth century, was one of the major waterworks of the Vijayanagara empire, near its capital at Hampi.

Q.23 · Answer (A) disguised unemployment
In disguised unemployment more people are engaged in work than it needs, so taking some away would not reduce output.

Q.24 · Answer (D) Three points
A basket scored from beyond the three-point line counts for three points. (Under FIBA's rules the line was 6.25 m from the basket until 2010; it is now 6.75 m.)

Q.25 · Answer (A) To provide financial services, such as accepting deposits, lending money and facilitating payments
Commercial banks accept deposits, lend money and handle payments. Issuing currency, setting policy rates and regulating the money supply are the work of the central bank (the RBI in India).

English Language and Comprehension

Q.1 · Answer (A) Mississippi
The correct spelling is 'Mississippi', whose delta is the classic bird's-foot delta.

Q.2 · Answer (B) Ophidiophobia
Ophidiophobia is the fear of snakes; xenophobia is fear of foreigners, necrophobia of death and nomophobia of being without a mobile phone.

Q.3 · Answer (C) region; produced
Cuba is part of the Caribbean 'region', and as the sugar bowl of the world it 'produced' a great deal of sugar.

Q.4 · Answer (C) completed
'To' as part of an infinitive takes the base form of the verb: 'determined to complete the project'.

Q.5 · Answer (A) proportion
The correct spelling is 'proportion', a part or share of a whole.

Q.6 · Answer (A) The responsibility was given to each student to keep quiet during the seminar.
'Keep quiet' takes the plain adjective, since nothing is being compared; the other sentences use 'quieter', and one also has 'was give' for 'was given'.

Q.7 · Answer (A) A season
Going to the Himalayas 'this spring' refers to the season; the other meanings of 'spring' do not fit.

Q.8 · Answer (A) Assesment
The correct spelling is 'Assessment'.

Q.9 · Answer (D) new bicycle confident
An adverb is needed to describe how she was riding: 'riding her new bicycle confidently'.

Q.10 · Answer (A) We came in the meeting room quietly.
Movement to the inside of a place is 'into': 'We came into the meeting room quietly.' The others use 'come across', 'come up with' and 'come down with' correctly.

Q.11 · Answer (C) wearing
Contact lenses are worn: 'How long have you been wearing these contact lenses?'

Q.12 · Answer (B) best
'Best' is already the superlative of 'good'; 'most best', 'more better' and 'most good' are wrong forms.

Q.13 · Answer (B) to
'Near' is itself the preposition, so 'to' is redundant: 'They live near the park.'

Q.14 · Answer (C) tongue in cheek
Something said 'tongue in cheek' is said ironically or as a joke, not meant seriously.

Q.15 · Answer (A) Enemy
An 'ally' is a friend or supporter; its opposite is an 'enemy'.

Q.16 · Answer (C) hardworking
'Diligent' means careful and hardworking.

Q.17 · Answer (C) set out
To 'set out' is to begin a journey; to 'set apart' is to make different from others.

Q.18 · Answer (D) Plot
A 'conspiracy' is a secret plan to do harm: a 'plot'.

Q.19 · Answer (D) Supported
To have 'agreed with' an assessment is to have shared and backed it: 'supported' it.

Q.20 · Answer (C) a show of hands
A 'show of hands' is a vote taken by raising hands.

Q.21 · Answer (B) spreads
A weed 'spreads' where it is not wanted.

Q.22 · Answer (C) dangerous
Most weeds are 'dangerous', that is harmful, as the next sentence shows: toxic to animals and harmful to crops. The other options are nouns.

Q.23 · Answer (D) depriving
To 'deprive' something of a thing is to take it away: weeds harm crops by 'depriving them of water'.

Q.24 · Answer (A) propagate
Weeds 'propagate', that is spread and multiply, in various ways; 'extension' is a noun, and 'entertain' and 'breach' make no sense here.

Q.25 · Answer (D) frequently
The adverb 'frequently' modifies 'transported': weeds are often carried from place to place.

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